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import gymnasium as gym
import numpy as np

def random_agent(env_name, episodes=10):

    print("\n" + "=" * 50)
    print("Environment:", env_name)
    print("=" * 50)

    # 1. Create the environment
    env = gym.make(env_name)

    rewards = []

    for episode in range(1, episodes + 1):

        # 2. Reset the environment
        state, info = env.reset()

        # 3. Observe the initial state
        print("\nEpisode:", episode)
        print("Initial State:", state)

        total_reward = 0
        terminated = False
        truncated = False
        steps = 0

        # 4-8. Select random actions until episode terminates
        while not (terminated or truncated):

            # Identify available actions
            action = env.action_space.sample()

            # Execute action
            next_state, reward, terminated, truncated, info = env.step(action)

            # Observe next state and reward
            state = next_state
            total_reward += reward
            steps += 1

        # 9. Display total reward
        print("Steps:", steps)
        print("Total Reward:", total_reward)
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        rewards.append(total_reward)

# Calculate average reward
average_reward = np.mean(rewards)

print("\n" + "-" * 50)
print("Results for", env_name)
print("-" * 50)
print("Total Rewards:", rewards)
print("Average Reward:", average_reward)

env.close()

# =====
# CliffWalking Environment
# =====
random_agent("CliffWalking-v1", episodes=10)

# =====
# Taxi Environment
# =====
random_agent("Taxi-v4", episodes=10)

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Environment: Taxi-v4
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Episode: 1
Initial State: 32
Steps: 200
Total Reward: -767

Episode: 2
Initial State: 303
Steps: 200
Total Reward: -749

Episode: 3
Initial State: 113
Steps: 200
Total Reward: -722

Episode: 4
Initial State: 373
Steps: 36
Total Reward: -168

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Episode: 5
Initial State: 448
Steps: 200
Total Reward: -695

Episode: 6
Initial State: 332
Steps: 200
Total Reward: -731

Episode: 7
Initial State: 394
Steps: 200
Total Reward: -704

Episode: 8
Initial State: 313
Steps: 200
Total Reward: -758

Episode: 9
Initial State: 286
Steps: 199
Total Reward: -781

Episode: 10
Initial State: 43
Steps: 200
Total Reward: -812

Results for Taxi-v4

Total Rewards: [767 740 722 160 605 721 704 758 712]

